

## Functions and Classes

1. Create a class Student with attributes name, roll\_number, and grades. Implement methods to add a grade, calculate the average grade, and display the student's details.
2. Write a function factorial(n) that calculates the factorial of a given number using recursion. Test it for n = 5.
3. Design a class Rectangle with attributes length and width. Add methods to calculate the area and perimeter. Also, include a method to check if the rectangle is a square.
4. Create a class Car with attributes make, model, and year. Add a method start\_engine() that prints a message indicating the car is starting.
5. Write a function fibonacci(n) that generates the first n numbers in the Fibonacci sequence.
6. Develop a class Book with attributes title, author, and price. Add methods to apply a discount to the price and display book details.
7. Write a program that defines a class Circle with methods to calculate the area and circumference based on its radius.
8. Create a function reverse\_list(lst) that reverses the elements of a given list lst without using built-in methods like reverse ().
9. Design a class BankAccount with methods for deposit, withdrawal, and checking the account balance.